Ch 3.3: The Last of the Linear Regression

Lecture 8 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Jan 30, 2026

Announcements

Last time:

- Started 3.3 Questions of linear regression

Announcements:

- HW #2 Due Sunday 2/8

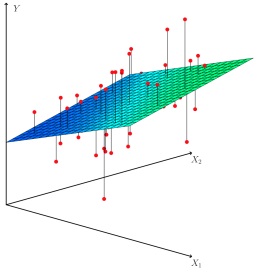
Covered in this lecture

- Extending the linear model with interaction terms
- Hierarchy principle
- Polynomial regression

Section 1

Review from last time

Linear Regression with Multiple Variables



- Predict Y on a multiple variables X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p x_p + \varepsilon$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1, \cdots$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

Section 2

Categorical Input Variables

Reminder: Qualitative vs Quantitative predictors

Copied from day 7 b/c out of time. can comment out!

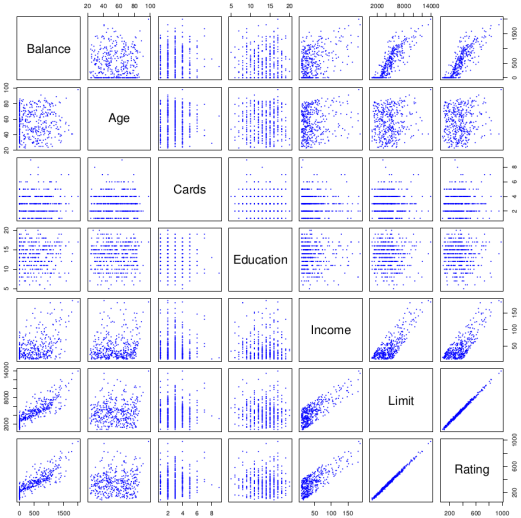
Quantitative:

- Anything with values in either integers or real numbers
- Temperature
- Year (even though integers)
- Weight
- Number of cylinders in a car

Qualitative/Categorical:

- Anything from an unordered set
- Country of origin
- Brand names
- Anything yes/no (married, has a cat)

Credit card balance



- `own`: house ownership *This is referenced in the text, but isn't in any of the online versions of the data set that I can find*
- `student`: student status
- `status`: marital status
- `region`: East, West, or South *Also referenced in the book, but not in the online data sets*

What if....

Copied from day 7 b/c out of time. can comment out!

... your variables aren't quantitative?

- Home ownership
- Student status
- Major
- Gender
- Ethnicity
- Country of origin

Example

Investigate differences in credit card balance between people who own a house and those who don't, ignoring the other variables.

One-hot encoding of categorical input variable

Create a new variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

- this is often called a dummy variable
- β_0 is avg credit card balance among those who are not a student
- $\beta_0 + \beta_1$ is the avg for those are
- β_1 is avg diff in balance

Interpretation

Copied from day 7 b/c out of time. can comment out!

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	480.3694	23.434	20.499	0.000	434.300	526.439
Student[T.Yes]	396.4556	74.104	5.350	0.000	250.771	542.140

- Avg CC balance for non-students \$480.36,
- Students have \$396.46 additional debt, so \$876.83

Model:

$$y = 480.36 + 396.46 \cdot x_{student}$$

Test your understanding: PollEv

Who cares about 0/1?

Old version: 0/1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Alternative version: ± 1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ -1 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Note that the predictions will be the same

Qualitiative Predictor with More than Two Levels

`Region`:

	x_{i1}	x_{i2}
South	*1*	*0*
West	*0*	*1*
East	*0*	*0*

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases} \end{aligned}$$

More on multiple levels

- Always one less variable than levels
- Level with no dummy variable is called the baseline

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	531.00	46.32	11.464	< 0.0001
`region[South]`	-18.69	65.02	-0.287	0.7740
`region[West]`	-12.50	56.68	-0.221	0.8260

- Balance for the baseline (EAST) is \$531
- South has \$18.69 less than east
- West has \$12.50 less than east
- p-values for dummy variables are large, so can't reject null hypothesis
- No statistical evidence that there's a diff in CC balance in different regions

Do code section on "Dummy Variables for Multi-level Categorical Inputs"

Section 3

Extending the linear model

Assumptions so far

Back to our Advertising data set

$$\hat{Y}_{sales} = \beta_0 + \beta_1 \cdot X_{TV} + \beta_2 \cdot X_{radio} + \beta_3 \cdot X_{newspaper}$$

Assumed (implicitly) that the effect on sales by increasing one medium is independent of the others.

What if spending money on radio advertising increases the effectiveness of TV advertising? How do we model it?

Interaction Term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \\ &= \beta_0 + \tilde{\beta}_1 X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

- $\beta_3 X_{radio} X_{TV}$ is the interaction term
- $\tilde{\beta}_1 = \beta_1 + \beta_3 X_{radio}$, function of X_2 association no longer constant between X_1 and Y

Interaction term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	6.7502	0.248	27.23	< 0.0001
`TV`	0.0191	0.002	12.70	< 0.0001
`radio`	0.0289	0.009	3.24	0.0014
`TV×radio`	0.0011	0.000	20.73	< 0.0001

- P-value for interaction term is low, strong evidance for $H_a : \beta_3 \neq 0$.
- R^2 is 96.8% for this, vs 89.7% for model without interaction
- β_3 the increase in the effectiveness of TV advertising associated with a one-unit increase in radio advertising

Interpretation

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- Increase in \$1K of TV is associated to increased sales of

$$(\beta_1 + \beta_3 X_{radio}) \cdot 1000 = 19 + 1.1 \cdot X_{radio}$$

units

- Increase in radio advertising of \$1K will be associated with an increase sales of

$$(\beta_2 + \beta_3 X_{TV}) \cdot 1000 = 29 + 1.1 \cdot X_{TV}$$

Hierarchy principle

Sometimes p-value for interaction term is very small, but associated main effects are not.

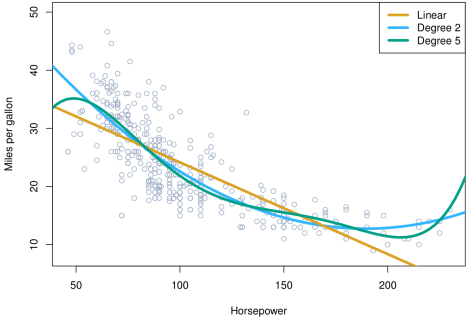
The hierarchy principle:

- If we include the interaction term, we should include the main effects no matter what.
-

Nonlinear relationships

$$\texttt{mpg} = \beta_0 + \beta_1 \cdot \texttt{horsepower} + \beta_2 \cdot \texttt{horsepower}^2 + \varepsilon$$

- Quadratic shape of data suggests above model
- Still linear because we're doing multi-linear regression with $X_1 = \textit{horsepower}$ and $X_2 = \textit{horsepower}^2$
- Could add more terms, but risk of overfitting
- Polynomial regression, more in ch 7



	Coefficient	Std. error	t-statistic	p-value
`Intercept`	56.9001	1.8004	31.6	< 0.0001
`horsepower`	−0.4662	0.0311	−15.0	< 0.0001
`horsepower`^{2}	0.0012	0.0001	10.1	< 0.0001

Do the section on “Interaction Terms”

Next time

CMSE381_S2026_Schedule : Sheet1

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression				
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	